A Precise Oriented Ship Detector in SAR Images Based on Dynamic Rotated Positive Sample Mining

Tingxuan Yue, Yanmei Zhang, Jin Wang, Yanbing Xu, Pengyun Liu, and Chengcheng Yu

***Abstract*—The synthetic aperture radar (SAR) oriented ship detection is a more complex and finer task than the horizontal, and it provides ship orientation for military and civilian applications, such as trajectory prediction. However, many oriented target detection methods discarded high-resolution (low-level) feature map, which provides morphological features for predicting accurate bounding box for small ships, due to computational cost and lack of semantic information. In addition, most methods follow the samples assignment in horizontal detection after padding oriented bounding box. The inaccurate geometric relationship causes that some low-quality positive samples without contributing accuracy improvement are introduced. To improve the precision of small oriented ship detection, we proposed the feature pyramid network contained high-resolution feature to fuse high-level semantics and highlight contours and textures of ships in the low-level feature map. To select high-quality positive rotated samples, dynamic rotated positive samples mining (DRPSM) is designed to mine hard positive samples for oriented ship detection specially. In one word, a two-stage and anchor-free detector trained by DRPSM and utilized high-resolution feature map is proposed for detecting oriented ships in SAR imagery. Subsequently, extensive experiments have been conducted on two oriented SAR datasets: the SSDD and the HRSID. As a result, our method attains recall of 96.2% and 90.1%, and AP50 of 90.7% and 86.5% on SSDD and HRSID, respectively. It shows that our detector achieves state-of-the-art results applied in capturing ships, when compared with other popular remote sensing oriented detectors.**

***Index Terms*—Convolution neural network (CNN), oriented detection, remote sensing, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) target detection is an important field of research in remote sensing [1]. Its all-weather observation advantage has led to a wide range application of military and civilian, particularly in maritime ships monitoring. With the development of SAR imaging technology, high-resolution SAR images are used increasingly for ship detection [2], [3].

Traditional SAR ship detection methods rely on manual feature extraction from targets [4] or statistical models of the sea surface [5]. The rapid development of deep learning techniques has significantly improved the accuracy of ship detection. The convolution neural networks are adapted from common objects detection to SAR ship detection. Zhu et al. [6] inserted spatial pyramid pooling into the architecture of YOLOv3 [7] for improving the accuracy of the ship detection. Some scholars improved the representation ability of the feature map generated by the network. The high-resolution and low-resolution convolutions were adopted to maintain and strengthen the representations of ships in feature maps respectively [8]. Lin et al. [9] further enhanced the representation of feature maps through shallow feature reconstruction. The attentional mechanisms modeled on human vision are also used to enhance ship-related information in feature maps. The coordinate attention module was embedded in backbone to make features represent ships better, and multiscale contextual information was captured by the network for detecting ships in SAR images [9], [10]. The attention guided balanced pyramid was proposed to solve the imbalance in multiscale features and reduce interference of irrelevant information in SAR ship detection [11]. In addition, some obstacles in SAR ship detection have also been alleviated. The adverse effects of fuzzy areas caused by SAR imaging mechanism have been reduced by guide position regression branch and redesigned regression method. The mosaic data augmentation was introduced to solve the limit of insufficient data in the stage of preprocess ship samples [6]. These structures and tactics designed for detecting ships can locate ships more accurately. However, these methods are still short for locating ships using the horizontal bounding box (HBB) in the detection of common objects (such as COCO dataset [12]). The HBB contains redundant areas unrelated to ships and may include the coast or adjacent ships as part of the target.

Orientation bounding box (OBB) can locate targets more closely to its shape compared with HBB, and more and more OBB detection methods are proposed for target detection in remote sensing [13]. Stepwise regression from an anchor to an OBB is a direct and effective approach for arbitrarily rotated objects. This route of the coarse-to-fine includes more than two detection heads (refinement stages), and the intersection over union (IoU) thresholds of stages increase gradually [14]. To predict orientation variation of targets, it is necessary to capture rotation-equivariant features. The ReDet [15] encodes equivariance and invariance of rotation, and then extracts rotation-invariant features according to the orientation region of interest (ROI). The coding method of OBB is also a worthy research entry point, which can reduce the difficulty of OBB prediction. Xu et al. [16] proposed an innovation framework to product

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OBB of oriented targets from its HBB, which regresses the gliding offset of four vertexes in HBB to obtain OBB. The ship detection method based on two-stage network usually predicts the region proposal as HBB, while the oriented R-CNN [17] generates oriented region proposals directly, which eliminates the step of refinement from HBB to OBB. Simultaneously, the redesign of loss function can effectively ease the problem of boundary discontinuity in the regression of OBB in optical remote sensing target detection [18], [19], [20]. Unfortunately, the lack of multichannel colour information in SAR imagery makes it difficult to apply OBB detection methods from optical remote sensing directly to ship detection in SAR imagery [3]. Compared with optical remote sensing imagery, ship detection in SAR imagery has fewer entry points to improve accuracy.

The OBB indicates the ship's orientation and provides information for complex sea surface monitoring tasks (such as ship trajectory prediction). Therefore, the methods of oriented SAR ship detection are becoming an important research hotspot. It is an effective way to adapt other domain OBB detection methods to SAR. The coarse-to-fine route in R3Det has been adopted in detecting oriented ships in SAR imagery, and enhancing contextual information and highlighting spatial features for feature maps from backbone can be used for regressing OBB of ships [21]. Guo et al. [22] utilized inclination from left-top point to obtain four OBB vertices, which can effectively break through the border restriction of HBB for OBB. This coding method is more suitable for SAR ship OBB detection compared with midpoint offsets in oriented R-CNN [17]. In addition, some scholars modified the common object detection method to obtain a SAR ship OBB detection network. Zhao et al. [23] introduced three branches of ratio-ness, ship aspect angle, and rotation regression into the head of FCOS, and these branches regress and refine OBB for ship detection. Sun et al. [24] proposed bidirectional feature fusion module to aggregate multiscale features to detect multiscale ships, and they explored a more suitable angle representation method for SAR ship detection. The scattering characteristics are an unique entry point for detection in SAR imagery. Sun et al. [25] utilized a grouping network to obtain strong scattering points for generating embedding vectors, and the OBBs are predicted by these vectors, as well as calculate the corresponding minimum bounding rectangles by key points of objects. The scattering characteristics of SAR ship can also be combined with geometric features. Gao et al. [26] proposed ellipse encoding to encode the contour of scattering area as a set of the ordered key point distances and the angle intervals. The development of vision transformer model has inspired researchers. Zhou et al. [27] incorporated overlapping patch embedding and mixed transformer encoder modules to overcome the obstacle of pyramid vision transformer (PVT) model migrating from the natural image to SAR ship detection.

The above methods provided innovative approaches for applying neural network (NN) to SAR ship OBB detection. However, they all failed to fully exploit the information of the high-resolution feature map, which can lead to a mismatch between receptive fields and small ships. And the spatial information, such as texture and boundaries of the target in low-level features helps the network to locate the small ships [28] and regress OBB for ships accurately. In addition, the label assignment dominates the performance of the network. The HBB detector typically uses the horizontal IoU (HIoU) between the ground truth (GT) boxes and the predicted boxes as the metric for assigning positive samples in training. These methods of OBB detection still retain HIoU as assignment metric by padding the OBB as HBB, however, the HIoU cannot accurately represent the geometric relationship between the predicted OBB and the GT OBB, caused by padding extra parts. In this process, low quality positive samples are introduced into the training stage of the network.

To overcome the obstacle and further improve the accuracy, we proposed an anchor-free oriented SAR ship detector with FPNHRF and a sample assignment method named DRPSM. The detector follows the structure of two-stage network, but drops the design of preset anchor unlike other two-stage networks [29], and the region proposal network (RPN) generates rotated proposals straightforwardly. We proposed the FPNHRF to replace the feature pyramid network (FPN) [30] to capture multiscale ships. The FPNHRF adds high-resolution feature map C_1 to the FPN, i.e., the feature map C_1 output from the ResNet is the first input of the FPN. The C_1 fully preserves textures and edges of ships but lacks semantic information, and the receptive field of C_1 can match the small ships. Therefore, we extended the top-down pathway of FPN from P_2 to C_1 thus enriching the semantic information of C_1. Then, the global attention mask module (GAMM) is used to fully utilize global context information to enhance the relevant information and inhibit the irrelevant information in the low-level feature map of C_1. The quality of the positive samples determines the trend of the network training. The DRPSM mines hard samples, combining the midpoint offset representation [17] to generate accurate rotated proposals in this article. First, the DRPSM calculates the number of candidate positive rotated samples by HIoU between samples and GT boxes, both padded as HBB. The candidates are selected according to a cost including center metric and RIoU, which can accurately measure geometric relationship. Subsequently, high-quality positive samples are filtered from candidates depended on their loss calculated with GT boxes. The number of positive rotated samples is also dynamic, like candidates, calculated by RIoU between samples and GT boxes. In the second stage, the predicted OBBs of ships are refined based on rotated proposals. The major contributions of our work can be summarized as follows.

1) We proposed a two-stage oriented ship detector without preset anchor. The oriented proposals are regressed by anchor points rather than anchors in rotated RPN (RRPN).
2) To improve the precision of small oriented ship detection, the FPNHRF has been proposed for addressing the mismatch between receptive field and small ships. The GAMM we proposed can enhance relevant information of P_1 in FPNHRF by fully utilizing global context information.
3) The DRPSM has been proposed in this article to select a dynamic number of high-quality positive rotated samples. The cost in the DRPSM measures the geometric relationship between samples and GT accurately, and high-quality positive samples are mined depended on the loss function.

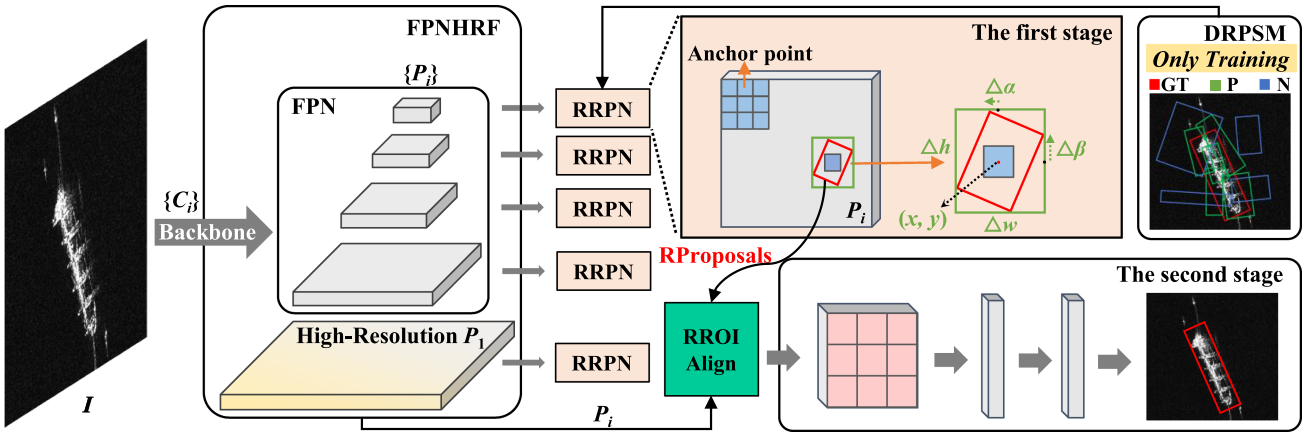


Fig. 1. Overall architecture of our method. The DRPSM assigns positive and negative samples only during the training, where GT, P, and N represent GT box, positive samples and negative samples, respectively. The RProposals and RROI represent rotated proposals and RROI, respectively.

4) To verify the effectiveness of our method, we conducted extensive experiments on two widely used SAR image ship datasets: SSDD [31] and HRSID [32]. Our method attains R of 96.2% and 90.1%, AP_{50} of 90.7% and 86.5% on SSDD and HRSID, respectively.

The rest of this article is organized as follows. Section II presents the details of our method. Section III contains the analyzation about the effectiveness of our method with experimental data, and Sections IV describes a discussion. Finally, Section V concludes this article.

II. Proposed Method

This section describes the proposed two-stage detector in detail. The overall architecture of our method is shown as Fig. 1. The feature maps $\{C_i, i = 1, \ldots, 5\}$ are extracted from the SAR imagery I. Different from other methods, we retain the C_1 with high resolution and fed all of $\{C_i, i = 1, \ldots, 5\}$ into the FPNHRF. The lack of semantic information of low-level feature [30] has been solved by the FPNHRF. And high-resolution feature map P_1 is appended to $\{P_i, i = 2, \ldots, 5\}$ to input into the RRPN. To improve the inferencing speed and generalizability of the detector, we use only one square anchor called anchor point to replace the multiple preset anchors in other anchor-based methods. We also introduce the midpoint offset representation in oriented R-CNN [17] to regress more accurate rotated proposals. Therefore, the RRPN predicts the center, width, and height of the horizontal anchor from the anchor point, as well as the offsets of midpoint of edges in the horizontal anchor to generate rotated proposal. The majority of negative samples are generated in FPN, while a minority of positive samples are required to balance the training process. The DRPSM mines high-quality rotated positive samples for training of RRPN. The rotated RoI (RROI) align corresponds to the rotated proposal to extract feature from P_i. More accurate OBBs are refined on the basis of the rotated proposals in the second stage.

A. Basic Framework

Our OBB oriented ships detection network is similar in structure to the HBB-labeled ship detection networks. The OBB detection needs to extract the finer features than HBB detection and predict angle additionally. We use the long edge definition 90° as definition of OBB in this article. The representation of OBB is (x, y, w, h, α), where (x, y), (w, h), and α are center localization, shape (width, height), and angle of OBB, respectively.

The ReDet [15] and ROI transformer [33] both follow the idea of regression from horizontal proposals to OBB. Different with both methods, our method generates rotated proposals directly and consequently refines to OBBs. The midpoint offset representation is introduced into RRPN to code rotated proposals. The oriented bounding box $(x, y, \Delta w, \Delta h, \Delta\alpha, \Delta\beta)$ is decoded by $\sigma = \{\sigma_x, \sigma_y, \sigma_w, \sigma_h, \sigma_\alpha, \sigma_\beta\}$ based on the anchor point (a_x, a_y, a_w, a_h) as (1). The $(x, y, \Delta w, \Delta h)$ is the external horizontal rectangle of the oriented bounding box as shown in Fig. 1, where (x, y) and $(\Delta w, \Delta h)$ are center localization and (width, height). The $\Delta\alpha$ and $\Delta\beta$ are the offsets of midpoints of width and height sides in horizontal rectangle $(x, y, \Delta w, \Delta h)$.

$$\begin{cases} \Delta\alpha = \sigma_\alpha \cdot \Delta w, \quad \Delta\beta = \sigma_\beta \cdot \Delta h \\ \Delta w = a_w \cdot e^{\sigma_w}, \quad \Delta h = a_h \cdot e^{\sigma_h} \\ x = \sigma_x \cdot a_w + a_x, \quad y = \sigma_y \cdot a_h + a_y. \end{cases} \tag{1}$$

The vertexes (v_1, v_2, v_3, v_4) of parallelogram obtained by the offset midpoint can be expressed as follows:

$$\begin{cases} v_1 = (x, y - \Delta h/2) + (\Delta\alpha, 0) \\ v_2 = (x + \Delta w/2, y) + (0, \Delta\beta) \\ v_3 = (x, y + \Delta h/2) + (-\Delta\alpha, 0) \\ v_4 = (x - \Delta w/2, y) + (0, -\Delta\beta). \end{cases} \tag{2}$$

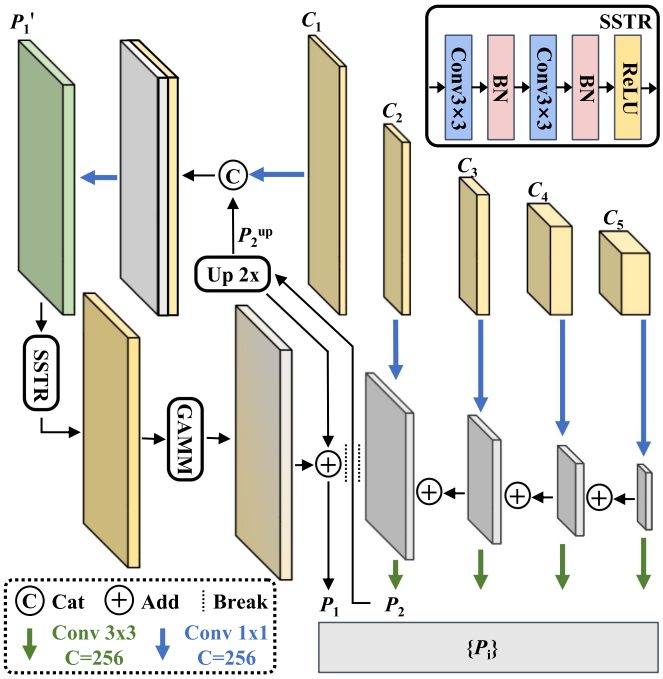


Fig. 2. Architecture of FPNHRF. The "Up 2×" represents the operation of double up-sampling.

Then, the rotated proposal (x, y, w, h, α) is converted from parallelogram (v_1, v_2, v_3, v_4) as follows [34]:

$$\begin{cases} (x, y) = (v_1 + v_3)/2 \\ \text{edge1} = \sqrt{\sum (v_1 - v_2)^2} \\ \text{edge2} = \sqrt{\sum (v_2 - v_3)^2} \\ w = \max(\text{edge1}, \text{edge2}), h = \min(edge1, \text{edge2}) \\ \alpha = \begin{cases} atan2(v_2 - v_1), \text{edge1} > \text{edge2} \\ atan2(v_4 - v_1), \text{edge1} <= \text{edge2} \end{cases} \end{cases}, \quad (3)$$

where $atan2(a - b)$ is to calculate angle of a line from two points a, b on the line. The transformation method is rotating the short sides around their midpoint to the direction of the vertical long sides to obtain the vertexes of the rectangle with direction, and it is different from the method in oriented R-CNN.

Subsequently the fixed-size feature ($7 \times 7 \times C$, C is the number of channels) is extracted by RROI align. The rotated proposal is projected on the P_i to obtain RROI. Each RROI is divided into 7×7 grids, and the part of the fixed-size feature, which corresponds on each gird can be calculated by

$$\sum_{(x,y) \in gird} P_i(T(x, y, \alpha))/n \quad (4)$$

where $T(\cdot)$ is rotated transformation from [33], and n is the number of sample points in the gird.

B. FPNHRF

The multiscale detection and semantic fusion proposed in FPN [30] significantly improve the accuracy of object detection. $\{P_i, i = 2, \ldots, 5\}$ in Fig. 2 generate by FPN as follows:

$$P_i = \begin{cases} \text{Conv}_{3\times3}(\text{Conv}_{1\times1}(C_i) \\ \quad +\text{Upsample}(\text{Conv}_{1\times1}(C_{i+1}))), i = 2, \ldots, 4 \\ \text{Conv}_{3\times3}(\text{Conv}_{1\times1}(C_i)), i = 5 \end{cases}, \quad (5)$$

where $\text{Conv}_{1\times1}$ is laterals connection for channels reduction, and Upsample is the nearest interpolation. The $\text{Conv}_{a\times b}$ is the abbreviation of convolution, and the $(a \times b)$ is the size of convolution kernel. The resolution of feature map decreases as the level increasing in FPN, while the semantic information does the opposite. The low-level feature map has high resolution, but it is discarded due to lack of semantic information and large time cost. The LFCP [28] innovatively uses low-level feature map to improve precise of the ship instance segmentation. We proposed a structure based on the LFCP named FPNHRF, which is simpler and more suitable for ship detection compared with LFCP.

As shown in Fig. 2, the P_2 first undergoes a double up-sampling operation via CARAFE [35] as (6).

$$P_{2l}^{\text{up}} = \sum_{n=-2}^{2} \sum_{m=-2}^{2} W_{l(n,m)} \cdot P_{2(i+n,j+m)} \quad (6)$$

where P_{2l}^{up} means a location l in P_2^{up}, calculated by a reassembly kernel $W_{l(n,m)}$, which generates from P_2 and a square region on P_2 centered at (i, j). Each location l corresponds to a $W_{l(n,m)}$ and a square region.

$$P_1' = \text{Conv}_{1\times1}(\text{Cat}(\text{Conv}_{3\times3}(C_1), P_2^{\text{up}})) \quad (7)$$

where Cat is the abbreviation of concatnate.

We retained the SAR ship texture recovery (SSTR) block of LFCP to fuse semantic information from the higher levels. The SSTR block shown in top right of Fig. 2 processes the input feature F_{in} as follows:

$$F' = BN(\text{Conv}_{3\times3}(F_{\text{in}})) \quad (8)$$

$$F_{\text{out}} = \text{ReLU}(BN(\text{Conv}_{3\times3}(F'))). \quad (9)$$

where $ReLU$ and BN are $ReLU$ activation function and batch normalization. The reuse of global context block in LFCP is suitable for ship instance segmentation task but not suitable for oriented ship detection. First, global context block can capture long-range dependencies, which is very helpful for fine detection tasks. However, the cost of reusing the block is not worthy for oriented ship detection. In addition, we need to enhance the information related to ships as much as possible and reduce the effect of irrelevant information. The global context block focuses on capturing global context but may keep irrelevant information. Therefore, we designed a more simple module, the GAMM, to highlight the ship's edge and fully utilize global context information. The GAMM is shown as Fig. 3. For input feature $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, the GAMM can be formulated as follows:

$$Q = \text{GAP}(F_{\text{in}}), Q \in \mathbb{R}^{C\times1\times1} \quad (10)$$

$$K = \text{reshape}(F_{\text{in}}), K \in \mathbb{R}^{C\times HW} \quad (11)$$

$$M = \text{reshape}(\text{Softmax}(KQ^T)), M \in \mathbb{R}^{H\times W} \quad (12)$$

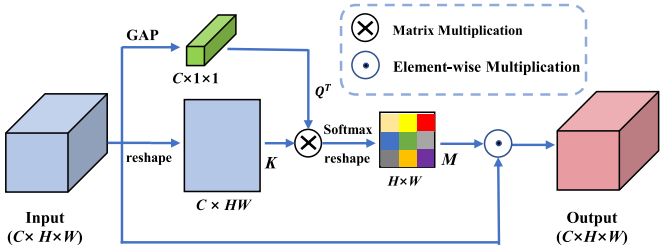


Fig. 3. Architecture of GAMM. The GAP represents operation of global average pooling.

$$F_{\text{out}} = F_{\text{in}} \odot M, F_{\text{out}} \in \mathbb{R}^{C\times H\times W} \tag{13}$$

where $GAP(\cdot)$ is global average pooling. The Q in GAMM is regarded as query of self-attention, and the self-attention models the spatial correlation between global feature representation and pixel vectors of feature map. The mask M represents similarity of pixel vectors of feature map, and the larger similarity indicates the greater relevance with ships. Then, the P_1 is obtained by the SSTR, the GAMM and skip connection in turn

$$P_1 = \text{GAMM}(\text{SSTR}(P_1')) + P_2^{\text{up}}. \tag{14}$$

To show the fact that the performance of our method does not depend on the preset anchor parameter, we tiled only square anchor at each position in a image and called it the anchor point. The basic scale of anchor point is robust, and the default setting is 4. The strides of anchor point are $\{2, 4, 8, 16, 32\}$ on $\{P_1, P_2, P_3, P_4, P_5\}$, respectively.

C. DRPSM

Due to the sparse distribution of ships on the sea surface, the number of negative samples is much higher than the number of positive samples in the RPN. Ma et al. [36] paid a huge calculation cost to obtain high-quality positive samples by presetting 54 R-anchor. The maximum IoU (HIoU) algorithm is widely used for positive and negative samples assignment in horizontal detection algorithms. However, the calculation of IoU has changed in oriented detection methods. Fewer positive samples are obtained at the beginning of training when the maximum RIoU algorithm is used directly. In addition, if the threshold of the maximum RIoU algorithm is adjusted to obtain more positive samples, low-quality positive samples will be introduced later in training. Therefore, directly replacing HIoU with RIoU in the maximum IoU algorithm is difficult to improve training accuracy.

The OHEM [37] mines the hard samples i.e., high-quality samples to reduce false positives. The SimOTA [38] calculates the number of positive samples assigned to each GT dynamically, and it improves the quality of positive samples by calculating the cost of the sample in combination with central prior knowledge. Unfortunately, existing sample assignment methods are mainly based on HIoU, and direct replacement of HIoU with RIoU will lead to degradation of the method performance. Some methods [15], [17], [29], [39] pad the OBB to HBB before applying the HBB sample assignment to select samples [34]. Fig. 4 shows the padding process from OBB to HBB (x_1, y_1, x_2, y_2)

Algorithm 1: Dynamic Rotated Positive Samples Mining.

Input:
$\mathbf{k}$ is a hyperparameter in the calculation of the number of samples selected;
$\mathbf{G}$ is a set of GT boxes on the image;
$\mathbf{A}$ is a set of anchor points;
$\mathbf{P}$ is a set of predicted bounding boxes corresponding to each element in $\mathbf{A}$;
$\mathbf{S}$ is a set of strides corresponding to each element in $\mathbf{A}$;
λ_1 and λ_2 are weight used for levelling of distances and loss of angle respectively.

Output:
$\mathbf{X}$ is a set of positive samples;
$\mathbf{N}$ is a set of negative samples.

for each GT $\mathbf{g} \in \mathbf{G}$ **do**
  compute RIoU between $\mathbf{P}$ and $\mathbf{g}$: $\mathbf{I}_R = RIoU(\mathbf{P}, \mathbf{g})$;
  padding OBB as HBB: $\mathbf{P}_H = padding(\mathbf{P}), \mathbf{g}_H = padding(\mathbf{g})$;
  compute HIoU between $\mathbf{P}$ and $\mathbf{g}$: $\mathbf{I}_H = HIoU(\mathbf{P}_H, \mathbf{g}_H)$;
  compute number of candidates: $n_c = Ceil(\sum_{top\,\mathbf{k}} \mathbf{I}_H)$;
  compute distances: $\mathbf{D} = \sqrt{\sum(\mathbf{P}_{center} - \mathbf{g}_{center})^2}/\mathbf{S}$;
  compute Cost: $\mathbf{Cost} = -\log(\mathbf{I}_R) + \lambda_1\mathbf{D}$;
  build an empty set for candidate positive samples of $\mathbf{g}$: $\mathbf{C} \leftarrow \varnothing$;
  $\mathbf{C} \leftarrow$ select n_c anchor points from $\mathbf{A}$ whose corresponding cost are the smallest n_c in $\mathbf{Cost}$;
  compute number of positive samples: $n = Floor(\sum_{top\,\mathbf{k}} \mathbf{I}_R)$;
  compute Loss between $\mathbf{C}$ and $\mathbf{g}$: $\mathbf{L} = L_{shape}(\mathbf{C}, \mathbf{g}) + \lambda_2|\mathbf{C}_{angle} - \mathbf{g}_{angle}|$;
  build an empty set for positive samples of $\mathbf{g}$: $\mathbf{X} \leftarrow \varnothing$;
  $\mathbf{X} \leftarrow$ select n anchor points from $\mathbf{C}$ whose corresponding loss are the biggest n in $\mathbf{L}$.
end for
$\mathbf{N} = \mathbf{A} - \mathbf{X}$
Return $\mathbf{X}$, $\mathbf{N}$.

like (15)

$$\text{padding}(x, y, w, h, \alpha) = \begin{cases} x_1 = x - \frac{1}{2}(w\cos\alpha + h\sin\alpha) \\ y_1 = y - \frac{1}{2}(w\sin\alpha + h\cos\alpha) \\ x_2 = x + \frac{1}{2}(w\cos\alpha + h\sin\alpha) \\ y_2 = y + \frac{1}{2}(w\sin\alpha + h\cos\alpha). \end{cases} \tag{15}$$

Areas not related to the ship are included in the padding bounding box, and low quality positive samples will be introduced into the training process. And the approach is inconsistent with the purpose of OBB detection.

We follow the purposes of ATSS [39] and dynamic soft label assigner [40]. 1) Explore an assignment of positive and negative samples, which is more suitable for oriented SAR ship detection. 2) Only tile a square anchor (anchor point) at each position of

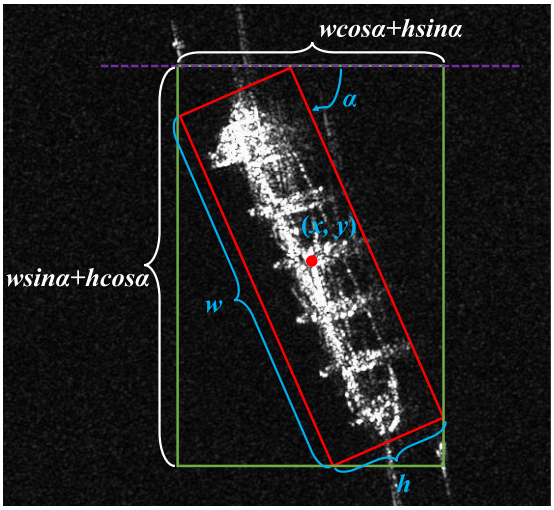


Fig. 4. Padding process from OBB to HBB. The red and green bounding boxes are original OBB and padded HBB, respectively. The purple dashed line is the horizontal line.

the image to avoid invalid calculation. 3) Dynamically adjust the number of positive samples assigned to each GT. The Algorithm 1 describes how proposed DRPSM selects positive and negative samples for an input SAR imagery in training. For each GT box g, we first select candidate positive samples, and the number of candidate positive samples is calculated dynamically by Line 3 to 5. The $Ceil(x)$ in Line 5 means that x is rounded up. The selection of candidates was based on central prior knowledge and RIoU as described in Line 2, 6, and 7, where the subscript $center$ means center localization of a bounding box. The candidate positive samples are obtained in Line 9. Hard positive samples i.e., high-quality positive samples tend to be harder to regress, that means a greater loss in candidates. Therefore, the positive samples are gathered from candidates according to Line 11–13, where the subscript $angle$ means angle of a OBB. The number of positive samples is also calculated dynamically in Line 10. The $Floor(x)$ in Line 10 means that x is rounded down. The numbers of candidates and positive samples are adjusted dynamically with the training by Line 5 and 10. The hyperparameter k will be analyzed in detail in the hyperparametric experiments. The λ_1 and λ_2 are weights used to ensure that each term in the equation contributes equally. The L_{shape} in Line 11 is calculated as follows:

$$L_{\text{shape}} = 2 - \min\left(\frac{w}{w_g}, \frac{w_g}{w}\right) - \min\left(\frac{h}{h_g}, \frac{h_g}{h}\right) \tag{16}$$

where the subscript g means the GT.

D. Others

1) Classification Branch: The branch of classification are omitted in Figs. 1 and 2. In the RRPN, a $\text{Conv}_{1\times1}$ is used for coarse ships classification as same as the faster R-CNN. The classification branch in the second stage shares two fully connected layers with the regression branch.

2) Loss: We choose the widely used cross entropy loss and smooth L1 loss as losses of classification(L_{cls}) and regression (L_{reg}), respectively, where smooth L1 is as follows:

$$\text{smooth}L1(x) = \begin{cases} 0.5x^2 & if\ |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases}. \tag{17}$$

Different from HBB detection, the OBB detection contains angle branches, so the parameters for inputting smooth L1 are (x, y, w, h, α). Our final loss function ($Loss$) for an SAR image as follows:

$$\text{Loss}(p_i, t_i) = \frac{1}{N_{\text{cls}}} \sum_i L_{\text{cls}}(p_i, p_i^*) + \frac{1}{N_{\text{reg}}} \sum_i p_i^* L_{\text{reg}}(t_i, t_i^*). \tag{18}$$

Here, p_i and t_i are the predicted probability and predicted bounding box of sample i, respectively, and i is the index of sample in a mini-batch. The p_i^* and t_i^* are GT label and GT OBB corresponding to sample i. The p_i^* is 1 if the sample i is positive, and p_i^* is 0 if the sample i is negative. The L_{cls} and L_{reg} are normalized by N_{cls} and N_{reg}, respectively. The N_{cls} is the number of mini-batch, and the N_{reg} is the number of the positive samples.

3) Hyperparameters Setting: The k in DRPSM is 13 and studied in hyperparameter experiment. The λ_1 and λ_2 sets 3.0 and 0.05 through experience, respectively.

III. Experiments

For describing the performance of our detector, we perform extensive experiments on a computer with NVIDIA RTX 3090 GPU. The compute runs ubuntu 20.04 OS and NVIDIA CUDA 11.0 is installed. The methods are implemented based on a toolbox developed specifically for rotation detection called MMRotate [34], which uses the Pytorch framework.

A. Datasets and Settings

With the development of SAR technology, many satellites have provided data with a resolution coverage of 0.5–15 m for SAR ship detection, such as TanDEM, RadarSat-2, TerraSAR-X, and Sentinel-1. The SSDD and HRSID are multisource and multiresolution SAR ship datasets produced from these data. There are 2456 ships in 1160 samples of SSDD, and the side length of samples is between 500 and 600 pixels. The sample size of the HRSID is 800×800 pixels, and the HRSID has a total of 5604 samples and 16951 ships of different sizes. The two datasets contain HBB annotations in the earlier open source version, and the OBB annotations of SSDD was published subsequently. We use the OBB annotations of HRSID provided by MMRotate. We divide the training set and the testing set of SSDD according to [31], i.e., 232 samples are used for testing and 928 samples are used for training. The sample assignment of HRSID corresponds to the [32], that is 3642 images and 1962 images in the training set and the testing set, and the ratio is 65% : 35%. In addition, we follow the setting of MMRotate: The images in the SSDD are resized to 608×608 pixels, and the images in the HRSID keep 800×800 pixels. The zero-padding operation is used during resizing samples for keeping the aspect ratio of the samples.

TABLE I
CONFIGURATION OF OPTIMIZER

Parameter	Value
Type	SGD
Initial learning rate	0.005
Epochs of learning rate decay	[11, 14]
Decay rate of learning rate	0.1
Momentum	0.9
Weight decay	0.0001

No samples augmentation method is used in our experiments. The checkpoint of backbone network pretrained on the ImageNet from the torchvision is loaded onto networks with ResNet50 as the backbone. The ReDet loads checkpoint of ReResNet onto backbone network at the beginning of training. All networks are trained on GPU with batchsize of 2 for 15 epochs. The configuration of the optimizer is shown in Table I. A linear warm-up strategy is applied to learning rate to reduce network instability in the early stages of training, and the number of warm-up iterations and the warm-up rate are 500 and 1.0/3, respectively. The NMS with a IoU threshold of 0.1 is adopted in the output of our detector in test. The rest of hyperparameters follow the default settings of the MMRotate Toolbox.

B. Evaluation Metrics

To objectively evaluate the performance of these methods, some quantitative metrics are adopt in this section, such as the most widely used recall (R) and precision (P). The precision-recall curve (PRC) is polt by (R, P), and shows their relationship specifically. And the OBB annotations of two datasets are both annotated in PASCAL VOC format [41]. Therefore, we introduce average precision with IoU threshold 0.5 AP_{50} from the VOC evaluation metrics to quantify the PRC instead of single P. The ability to capture ships is also vital for a detector, so R is used as an additional evaluation metric.

The R and P are defined by

$$R = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{19}$$

$$P = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{20}$$

where N_{TP}, N_{FN}, and N_{FP} are numbers of true positives (TP), false positives (FP), and false negatives (FN), respectively [3]. In oriented SAR ship detection, the TP is the correctly detected ship and the $RIoU >= 0.5$ between the OBB and GT box. The FP is a false alarm or the $RIoU < 0.5$ between the OBB predicted for the ship and GT box, and the FN stands for missed ships

$$\text{AP} = \int_0^1 P(R)\, dR. \tag{21}$$

The AP represents the area under the PRC. In addition, calculating the AP needs to set the RIoU threshold between the GT box and OBB predicted for the ship to determine the TP. The AP_{50} is the area under the PRC curve with a RIoU threshold of 0.5.

TABLE II
EXPERIMENTAL RESULTS OF HYPERPARAMETERS FOR k IN DRPSM

k	R (%)	AP_{50}(%)	Training time(s)	Parameters(M)
9	89.6	80.7	0.654	43.34
11	90.2	86.7	0.655	43.34
13	90.1	86.5	0.655	43.34
15	90.1	86.4	0.655	43.34
17	90.4	86.8	0.657	43.34

TABLE III
EXPERIMENTAL RESULTS OF HYPERPARAMETERS FOR SCALE (%)

Scale	R	AP_{50}
2	90.0	86.6
4	90.1	86.5

C. Hyperparameter Experiment

To investigate the effect of hyperparameters on the performance of our detector, we perform hyperparameter experiments on scale of anchor point and k in DRPSM. This experiment is conducted on the HRSID dataset, and setting of experiment is described above.

1) k in the DRPSM: Because the DRPSM mines hard positive samples, our detector can use the design of anchor point like the ATSS. Therefore, we explore the value of k to maximize the effectiveness of the DRPSM first. The experiment results of hyperparameters for k in DRPSM are shown in Table II. To ensure that some positive samples are assigned to the network at the start of training, we set $k \in \{9, 11, 13, 15, 17\}$. The R and AP_{50} are the lowest at k of 9, because the number of positive samples is calculated from the top k RIoU. Insufficient number of positive samples can influence the trend of training. The R and AP_{50} are the highest at k of 17, but the two metrics are approximately equal to other values of k in Table II. And the parameter sizes in Table II show that the DRPSM only effect the training stage. The results indicate that the hyperparameter k is robust, we choose k of 13 to balance training efficiency and indicator improvement.

2) Scale of Anchor Point: The scale, ratio of anchor, and even the angle of anchor in the rotation detection are very important hyperparameter in anchor-based methods. Our method does not require preset ratio and angle of anchor, but the scale is retained in order to use the midpoint representation. To verify that our method is insensitive to the scale hyperparameter, we choose commonly used scale values $\{2, 4\}$ for this experiment. The experiment result is shown in Table III, and the R and AP_{50} are differ by 0.1% when scale is 2 and 4. The performance of our method is insensitive to the value of scale, so our method can be considered as anchor-free. And the idea of ATSS that appropriate positive sample assignment methods can eliminate the need for predesigned anchors is also validated. We set the scale of anchor point to 4 as same as the ATSS.

D. Ablation Experiment

To verify the effectiveness of the FPNHRF and DRPSM in our method individually and jointly, the ablation experiment is conducted on the HRSID and SSDD dataset. Since our method

TABLE IV
RESULT OF ABLATION EXPERIMENT ON HRSID(%)

FPNHRF	DRPSM	R	AP_{50}
×	×	81.1	78.8
✓	×	88.9	80.5
×	✓	88.7	80.5
✓	✓	**90.1**	**86.5**

The bold values indicate the optimal indicator value among all the comparative models.

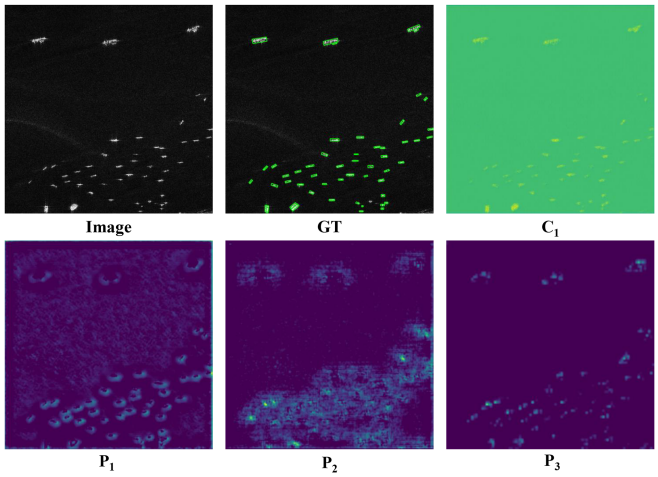


Fig. 5. Visualization of the feature maps in generating P_1 of FPNHRF. The Image is the original SAR image to be detected, and the GT is the ground truth. The C_1 is high-resolution feature map C_1 from the backbone network. The P_1 to P_3 are the top 3 low level feature maps in the FPNHRF.

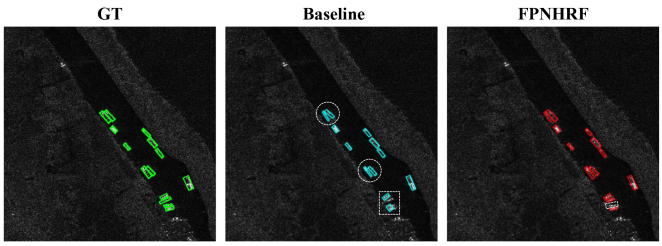


Fig. 6. Visualization results of the FPNHRF. The GT represents ground truth. The coloured rectangles are the detection results. The white rectangles and circles indicate regions of improvement after applying FPNHRF, and the circles correspond to the inaccurate bounding boxes and the rectangles correspond to missed ships.

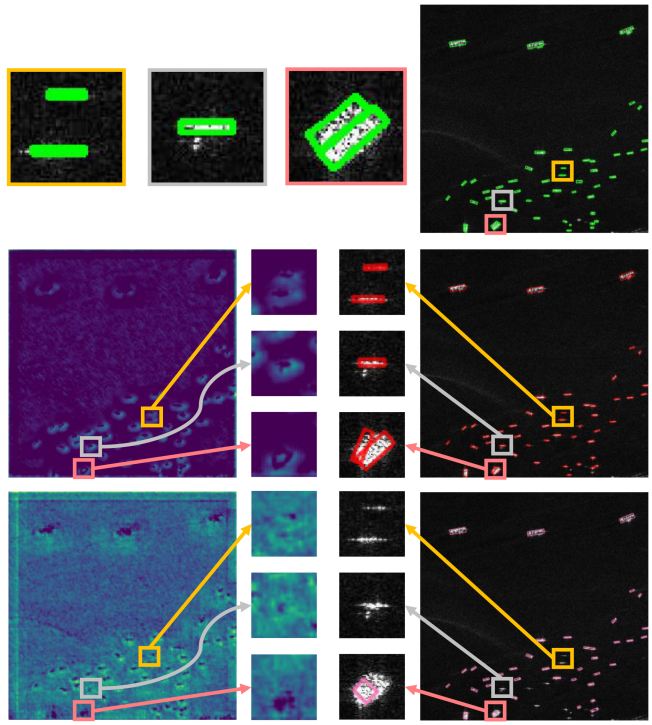


Fig. 7. Visualization of the lowest level feature map in FPNHRF and FPN. The top row is GT and the zoom-in target areas. The middle row is visual the lowest level feature map of FPNHRF, the zoom-in target areas on the feature map, the zoom-in detection result on image and the detection results of baseline with FPNHRF. The bottom row is visual the lowest level feature map of FPN, the zoom-in target areas on the feature map, the zoom-in detection result on image and the detection results of baseline.

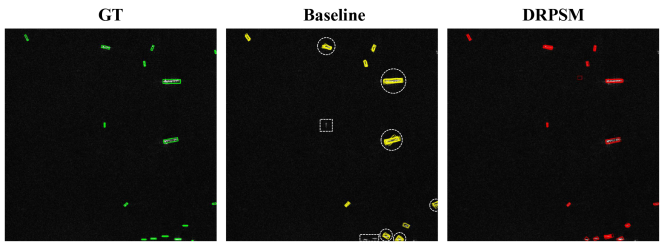


Fig. 8. Visualization results of the DRPSM. The GT represents ground truth. The yellow and red rectangles are the rotated proposals. The white rectangles and circles indicate regions of improvement after applying DRPSM, and the circles correspond to the inaccurate rotated proposals and the rectangles correspond to missed ships.

can be seen as having been developed on the basis of oriented R-CNN, we use it as the baseline for comparison. The result of baseline without FPNHRF and DRPSM is shown in the first row in Table IV.

The FPNHRF is designed for detecting small ships and predicting its angle of docking, and the effectiveness of FPNHRF is shown in second row in Table IV. The R and AP_{50} are improved 7.8% and 1.7%, respectively, and the significant improvement in R shows that the FPNHRF can reduce missed detection of ships effectively. This is because high-resolution feature map provides more information about small ships. And the improvement of AP_{50} indicates that morphology detail retained in high-resolution feature makes network regress more accurate OBB for ships. The C_1 in Fig. 5 shows the morphology detail retained, and the detail does not pay enough attention to the whole ships and has low contrast with the surrounding irrelevant information. The P_2 and P_3 are more focused on the whole ships, i.e., contains rich semantic information. The FPNHRF extends the semantic information from the higher level feature maps to C_1 to generate P_1, and the P_1 in Fig. 5 indicates that the high-resolution feature merges with semantic information. The scenario in Fig. 6 is a number of small ships docked on the shore, and interference from shore and neighbouring ships can easily lead detector to inaccurate predictions of OBB boundaries and missed ships. The baseline has problems with inaccurate OBBs of ship and missed ships as shown in the middle of Fig. 6. Fig. 6 shows that the OBBs of ship in white circles

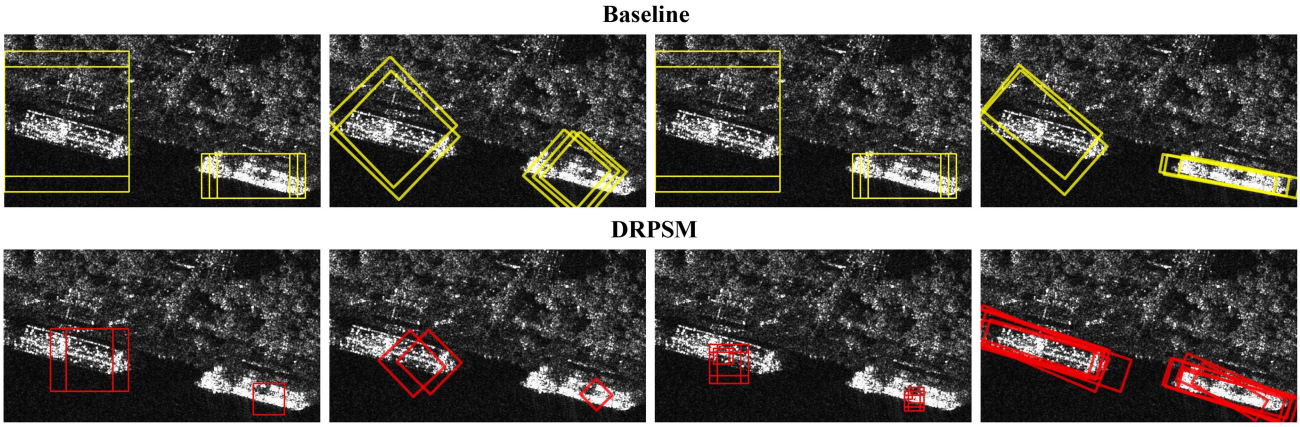


Fig. 9. Visualization of positive samples during training. The top row is the positive samples and corresponding anchors at the beginning and end of training for baseline. The bottom row is the positive samples and corresponding anchor points at the beginning and end of training for baseline with DRPSM. The second and first columns are the positive samples and corresponding anchors at the beginning training. The fourth and third columns are the positive samples and corresponding anchor points at the end training. There are two ships in the position of the positive samples concentration.

TABLE V
IMPROVEMENT PERFORMANCE OF FPNHRF ON SSDD

	$R(\%)$	$AP_{50}(\%)$	FPS(img/s)	Parameters(M)
LFCP [28]	95.1	90.6	13.6	43.42
FPNHRF	95.8	90.6	15.3	43.34

TABLE VI
COMPARISON OF TRAINING TIME ON HRSID (S)

Method	Time
Baseline	0.290
Baseline + ATSS	0.261
Baseline + DRPSM	0.305

region are more closer to the GT after applying the FPNHRF. And the number of missed ships in white rectangle region is reduced from 3 to 1 by the FPNHRF. Fig. 7 shows the comparison visualization of the lowest level feature map between FPNHRF and FPN. The contrast of the the lowest level feature map in the FPNHRF is higher than the one in the FPN, and the regional information is retained more in the FPNHRF. The small and blurred ships in the yellow rectangle can be detected due to the FPNHRF avoiding the mismatch between larger receptive fields and smaller ships. The contour and texture in P_1 of the FPNHRF helps the network capture ships with incomplete or unclear edges, such as the ships in gray and pink rectangles. In addition, the comparison experiment of time consumption between the FPNHRF and LFCP [28] is performed on the SSDD. The four metrics in Table V indicate that our GAMM is more suitable for ship oriented detection tasks than the reuse of global context block. Our FPNHRF captures more ships and obtains a faster inference speed compared with LFCP.

Positive sample assigner in the baseline is MaxIoUAssigner, which firstly pads both GT and predicted bounding boxes from OBB to HBB. Then, the MaxIoUAssigner using in HBB detection assigns samples for baseline. By using the DRPSM to replace MaxIoUAssigner, the R and AP_{50} are improved 7.6% and 1.7%, respectively, as shown in third row in Table IV. This is due to the fact that the high quality positive samples mined by DRPSM allow the network to detect more densely distributed and small ships. The scenario in Fig. 8 shows that small ships travelling intensively on the sea. Inaccurate rotated proposals and missed ships are shown in middle of Fig. 8, and the two obstacles caused by low quality positive samples in baseline can hinder the performance improvement of the network. The left in Fig. 8 demonstrates that the DRPSM has prompted RRPN to produce more accurate rotated proposals compared with MaxIoUAssigner. The proposals belonging to each GT in white circle regions are more similar in shape to GT, and the missed ships in rectangle region are matched to proposals. Fig. 9 shows the comparison of positive samples at the beginning and end of training between MaxIoUAssigner and DRPSM. The number of positive samples and corresponding anchors are invariant during training as shown in the top row of Fig. 9. The positive samples not contributing accuracy improvement for network are introduced in training caused by unbefitting anchors. On the contrary, the second row in Fig. 9 indicates the number of positive samples and corresponding anchor points are dynamically adjusted with network training. The number of positive samples is increasing as the bounding boxes predicted by the network are more and more accurate. The hard positive samples are mined by the DRPSM as shown in the fourth image of the second row in Fig. 9. These samples capture the ship but have a gap with GT, and they can prompt the network to generate OBBs more close to the GT. The DRPSM is more complex than MaxIoUAssigner, but there are fewer calculations involving anchors in DRPSM compared with MaxIoUAssigner. Therefore, we conduct the comparison experiment of training time on the HRSID, the results are shown in Table VI. The time in Table VI is obtained by averaging the training time of the network in the second epoch. The DRPSM only adds 0.015 s and 0.044 s of training time cost over the baseline and the ATSS, respectively.

Notably, the AP_{50} improves by 6% when the FPNHRF and DRPSM work jointly matched with individually, because adjacent and small ships are detected by a combination of high-resolution feature map and hard positive sample mining. There are strong clutter on the sea in Fig. 10, and the detection of the down left ship is interfered by the adjacent ship, and the

TABLE VII
COMPARISON WITH THE TYPICAL DETECTION METHODS ON SSDD(%)

	Method	Entire			Inshore		Offshore		FPS(img/s)	Parameters(M)
		R	AP_{50}	AP_{75}	R	AP_{50}	R	AP_{50}		
Single-stage	R-RetinaNet [44]	88.3	80.2	24.3	69.2	58.7	97.1	90.0	22.3	36.13
	CFA [42]	93.0	87.5	29.5	80.8	67.0	98.9	90.8	21.8	36.6
	CSL [18]	77.7	68.7	15.5	48.8	33.7	91.2	87.9	21.0	37.06
	R-Reppoint [45]	84.2	70.6	15.3	61.0	41.4	94.9	83.6	21.7	36.6
	GWD [20]	89.4	80.5	25.1	69.8	58.3	99.2	90.6	22.1	36.13
	KFIoU [19]	85.7	79.1	11.1	65.7	55.6	95.2	89.6	22.1	36.13
	R-ATSS [39]	92.9	90.1	27.1	79.7	70.1	98.9	90.8	22.5	36.01
	S2ANet [43]	90.5	89.1	31.8	72.1	66.8	98.9	90.8	17.9	38.54
Multi-stage	R-Faster-RCNN [29]	92.3	89.3	29.5	78.5	69.0	98.7	90.6	**27.9**	41.12
	Glidingvertex [16]	90.7	89.6	19.2	72.1	68.1	98.9	**90.9**	26.8	41.13
	Oriented-RCNN [17]	94.0	90.5	32.3	83.1	77.1	98.9	**90.9**	23.2	41.13
	R3Det [14]	91.2	89.6	47.7	75.6	69.7	98.4	90.7	16.5	41.58
	ReDet [15]	92.5	90.1	56.5	76.7	70.9	99.5	**90.9**	17.8	**31.54**
	RoI Transformer [33]	93.8	**90.7**	58.6	80.8	79.7	**99.7**	**90.9**	23.8	55.03
	Our Method	**96.2**	**90.7**	**67.8**	**89.5**	**81.1**	99.2	**90.9**	15.6	43.34

The bold values indicate the optimal indicator value among all the comparative models.

TABLE VIII
COMPARISON WITH THE TYPICAL DETECTION METHODS ON HRSID(%)

	Method	Entire			Inshore		Offshore		FPS(img/s)	Parameters(M)
		R	AP_{50}	AP_{75}	R	AP_{50}	R	AP_{50}		
Single-stage	R-RetinaNet [44]	79.2	69.1	34.9	59.6	41.8	94.8	89.6	19.5	36.13
	CFA [42]	82.4	74.8	39.3	64.7	44.5	96.5	90.4	18.0	36.6
	CSL [18]	73.8	66.1	25.2	50.8	34.4	92.1	88.6	17.8	37.06
	R-Reppoint [45]	79.2	64.4	27.0	60.2	36.7	94.2	86.5	18.3	36.6
	GWD [20]	80.3	72.1	36.0	60.9	42.6	95.7	89.8	19.1	36.13
	KFIoU [19]	75.3	67.3	32,8	50.0	32.2	95.3	89.7	18.7	36.13
	R-ATSS [39]	81.7	77.2	38.8	64.4	51.6	95.3	90.2	19.4	36.01
	S2ANet [43]	82.0	78.3	38.6	65.3	54.5	95.2	90.7	15.9	38.54
Multi-stage	R-Faster-RCNN [29]	81.0	78.0	44.1	62.9	54.7	95.4	90.1	**22.5**	41.12
	Glidingvertex [16]	78.8	71.3	36.4	59.9	48.5	93.8	90.4	22.2	41.13
	Oriented-RCNN [17]	81.1	78.8	50.1	62.8	55.4	95.7	**90.8**	21.4	41.13
	R3Det [14]	79.5	71.3	37.1	59.8	46.7	93.8	90.5	13.1	41.58
	ReDet [15]	82.6	79.2	48.6	65.5	55.5	96.1	90.7	14.3	**31.54**
	RoI Transformer [33]	84.0	79.7	49.3	69.7	56.8	96.7	90.7	19.8	55.03
	Our Method	**90.1**	**86.5**	**57.6**	**80.5**	**70.6**	**97.6**	90.7	9.4	43.34

The bold values indicate the optimal indicator value among all the comparative models.

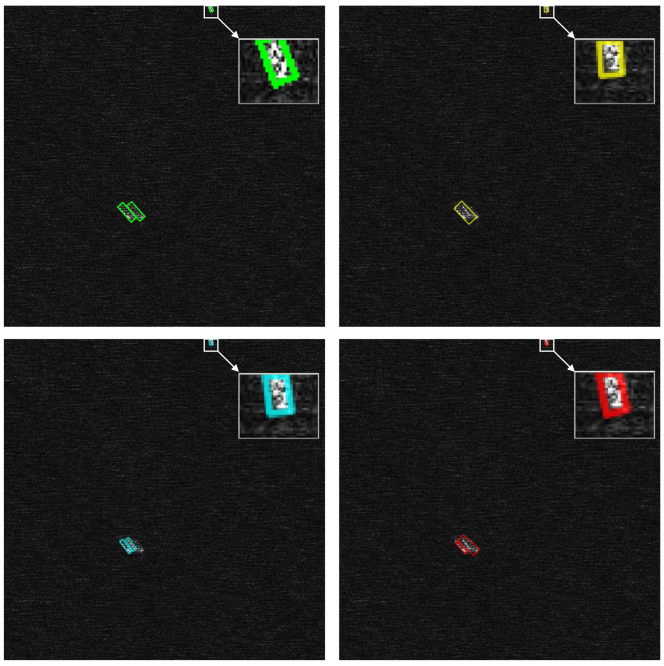


Fig. 10. Visualization results. The top left, top right, down left, and down right are results of GT, FPNHRF, DRPSM and our method, respectively.

top right ship is too small in scale. The FPNHRF in top right of Fig. 10 detects the boundaries of ships through high-resolution feature map, but it cannot distinguish the adjacent two ships due to their connected contours. The DRPSM assigns suitable positive samples to two adjacent ships resulting in predictions of OBB for the two ships. However, the OBBs have only bounded part of the ships because of the surrounding clutter. By taking advantages of both FPNHRF and DRPSM jointly, our method predicts more accurate OBBs for adjacent ships. In addition, the angle of the small ship submerged by clutter in the top right is regressed more closer to GT by our method.

E. Comparison

In this experiment, we compare the performance of our method with popular methods on the SSDD and the HRSID. Both HRSID and SSDD also provide the annotations of the inshore and the offshore, and we also test the R and AP_{50} on inshore and offshore test datasets for a more comprehensive comparison.

The methods with the prefix "R-" are modification of the horizontal detection method, such as R-RetinaNet, R-Faster-RCNN. The above two methods are the basis for developing other methods, and the "R-" means that the angle prediction

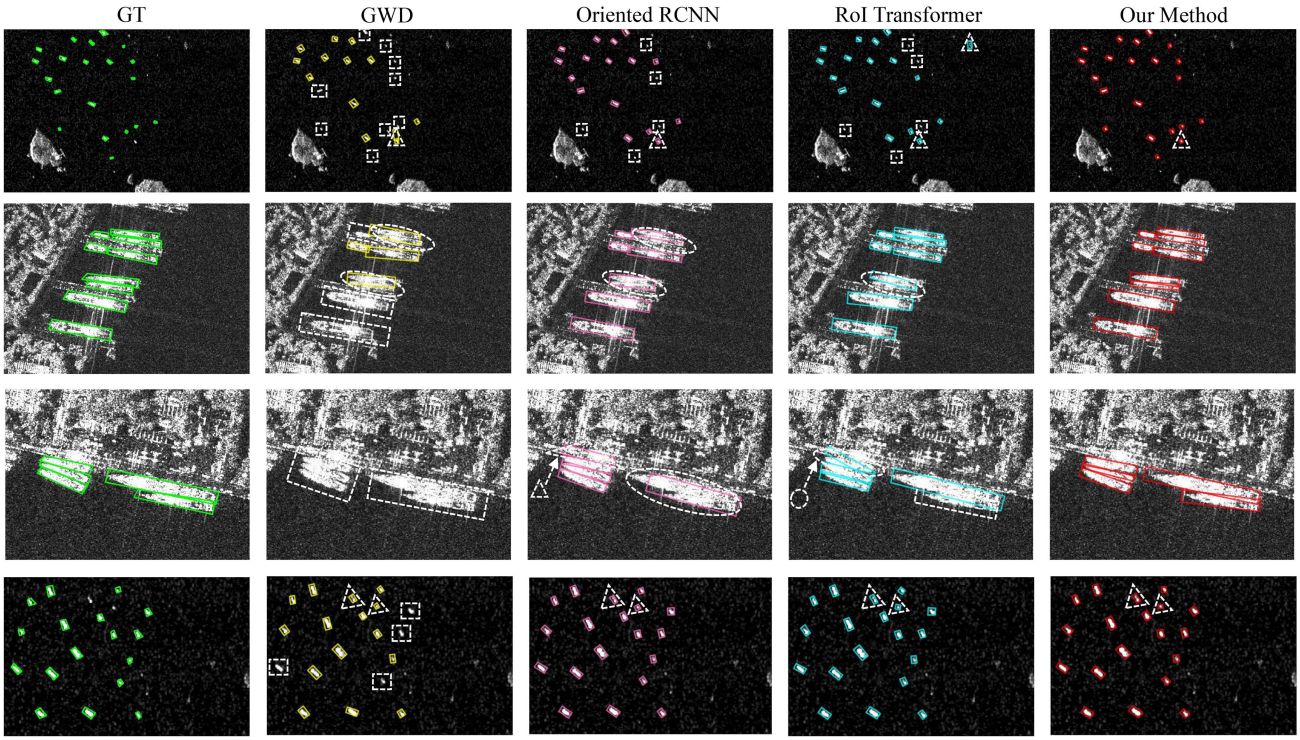


Fig. 11. Visualization results of four representative methods in typical and complex scenarios of the SSDD. The GT represents ground truth. The coloured rectangles are the detection results. The white rectangles, triangles, and circles represent the missing ships, the false alarm and inaccurate OBB, respectively.

branch is added to the detection head. The CSL [18], GWD [20], and KFIoU [19] focus on optimizing the loss function for oriented target detection. The CFA [42], Glidingvertex [16], and Oriented-RCNN [17] innovate the representation of the OBB. The S2ANet [43], R3Det [14], ReDet [15], and RoI transformer [33] have worked on angle prediction modules and feature alignment.

Experiment results are shown in Tables VII and VIII, and our detector obtains the highest R, AP_{50}, and AP_{75}, especially R which is over the 2nd 2.2% and 6.1% on entire test datasets of SSDD and HRSID, respectively. The highest AP_{50} and AP_{75} shows that our method locates the ship more accurately and the predicted OBBs by our detector fit the ship better. The highest R indicates that our approach can capture more ship targets, reducing the risk of missed alerts. This is due to small, easily ignored ships are detected by high resolution feature map in FPNHRF and ships disturbed by surrounding objects are regressed to more accurate OBB by mining hard positive samples through DRPSM. Our method also gain the highest R and AP_{50} on inshore test datasets of the SSDD and HRSID, and the metrics are higher than the 2nd method 6.4% and 1.4% on the SSDD, 10.8% and 13.8% on the HRSID. This means that the combination of the FPNHRF and the DRPSM can further overcome the effects of complex scenarios on detection results as shown in Fig. 10. The offshore ships are easy to be detected because of clear contours and edges, so most methods in Tables VII and VIII have similar metrics on offshore test datasets. In order to compare the performance of methods more comprehensively, three representative methods have been selected to visualize the comparison results. They are the GWD, the oriented R-CNN, and the RoI transformer, representing the three types of improvement, respectively, because they gain the best detection performance in its types of improvement as shown in Tables VII and VIII. The visualization results of four representative methods are shown in Figs. 11 and 12. We choose four typical and complex scenarios (dense distribution of small ships, adjacent docking of ships, inshore docking of ships, and interference from clutter and small islands) to compare the performance of the four methods. Missed ships and inaccurate OBB are common problems in detection scenario of dense distribution of small ships, and our method has the greater ability of detecting small ships as shown in the first row of Figs. 11 and 12. Adjacent and inshore docking of ships are detection challenges due to unclear edges and similar textures, and the other three compared methods all predict inaccurate OBB of ships as shown in the second and third row of Figs. 11 and 12. However, our method predicts more accurate OBB for these ships, which is consistent with the inshore results in Table VII and Fig. VIII. The forth row of Figs. 11 and 12 is a scene that shows small ships detection being disturbed by small islands and strong clutter. Our method detects all ships, while other methods all miss ships on the HRSID. These results show that our method is highly capable of detecting ships in complex detection scenarios.

In general, our method achieves a better detection performance than other popular remote sensing oriented detection methods, especially in capturing ships.

IV. Discussion

The R-ATSS and R-RetinaNet have the same network model, except that the former uses a more advanced samples assignment

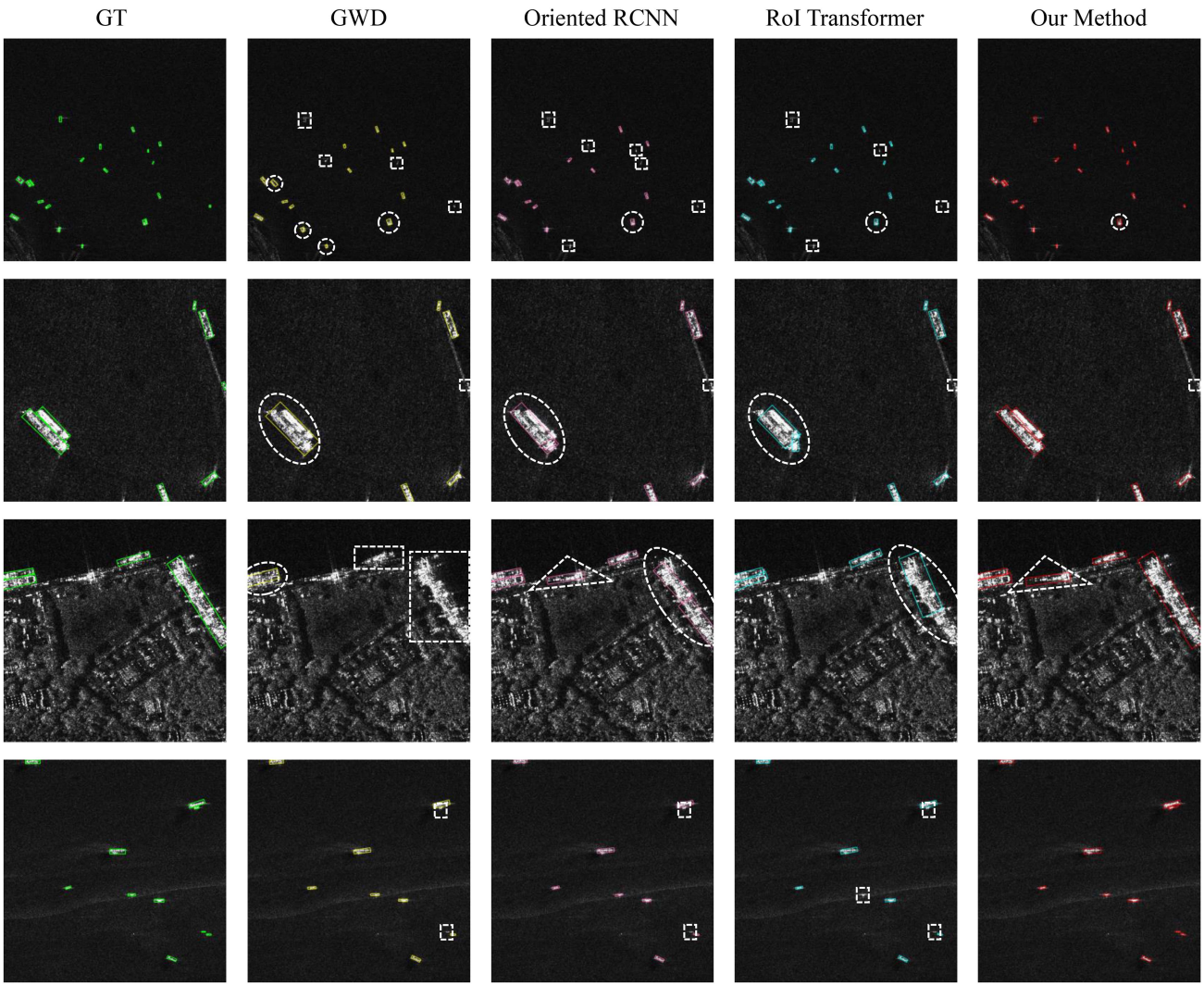


Fig. 12. Visualization results of four representative methods in typical and complex scenarios of the HRSID. The GT represents ground truth. The coloured rectangles are the detection results. The white rectangles, triangles, and circles represent the missing ships, the false alarm and inaccurate OBB, respectively.

strategy. However, the R-ATSS gains 4.6% and 2.5% of R improvement, and 9.9% and 8.1% of AP_{50} improvement over the R-RetinaNet on the two datasets, respectively. This means that a suitable sample assignment strategy can lead to the performance improvement without adding the network structure, and Table IV in ablation experiment also verifies the conclusion. Therefore, it is essential that we design the DRPSM specifically for oriented SAR ship detection. The threshold of calculating IoU is highly sensitive for metric AP, i.e., a slight shift of predicted box can result in a tremendous drop in its IoU with GT box. Therefore, the higher threshold of AP can better reflect the accuracy of predicted box, and a coarse predicted box can not meet the requirements of judging the ship direction and measuring the ship size. There are significant metrics AP_{75} of our method in Tables VII and VIII, and it means that the predicted boxes from our method bound ships more tightly compared with other methods. In addition, marine surveillance requires detection methods to minimize missed detections of ships. The strong ability to capture ships of our method connotes pretty practical value.

V. Conclusion

In this article, we proposed a two-stage detector utilizing high-resolution feature map and trained by DRPSM for detecting oriented ships in SAR imagery. Since there is no need to adjust the scale and preset ratio of anchors, the detector can be considered as anchor-free. For fully using morphological feature in low level feature map and addressing the mismatch between receptive fields and small ships, we proposed the FPNHRF to incorporate semantic information from higher levels, and the GAMM was inserted into FPNHRF to inhibit irrelevant information and highlight the information related to ships. Padding OBB to HBB is a common tactic of samples assignment in oriented target detection methods, and the inaccurate measurement of geometric relationships introduces low quality positive samples that do not contribute accuracy improvement in training. To overcome the obstacle, we designed the DRPSM to mine hard positive samples for oriented ship detection specially. The number of positive samples increases dynamically with the training process. Extensive experiments on the SSDD and HRSID show

that our detector achieves a better performance compared with other popular remote sensing detectors.

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